

Notice how the matrix in (3) is just the matrix of coefficients of the system (1). Similar calculations show that any system of linear equations, or any vector equation such as (2), can be written as an equivalent matrix equation in the form $Ax=b$.

Theorem 3: If A is an $m \times n$ matrix, with columns $a_1, \dots, a_n$, and if b is in $\mathbb{R}^m$, the matrix equation $Ax=b$ (4) has the same solution set as the vector equation $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$ (5) which, in turn, has the same solution set as the system of linear equations whose augmented matrix is $\begin{bmatrix} a_1 & a_2 & \dots & a_n & b \end{bmatrix}$ (6)

Existence of Solutions
The definition of $Ax=b$ leads directly to the following useful fact.
The equation $Ax=b$ has a solution if and only if b is a linear combination of the columns of A .

Section 1.3 considered the existence question, "Is b in $\text{Span}\{a_1, \dots, a_n\}$?" Equivalently, "Is $Ax=b$ consistent?" A harder existence problem is to determine whether the equation $Ax=b$ is consistent for all possible b .

Example 3: Let $A = \begin{bmatrix} 1 & 3 & 4 \\ -4 & 2 & -6 \\ -3 & -2 & -7 \end{bmatrix}$ and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$. Is the equation $Ax=b$ consistent for all possible b ?

Solution: The equation $Ax=b$ is consistent means that there exists a solution set of x . $\therefore b$ must be a linear combination of the columns of A , i.e. $x_1 \begin{bmatrix} 1 \\ -4 \\ -3 \end{bmatrix} + x_2 \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix} + x_3 \begin{bmatrix} 4 \\ -6 \\ -7 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$
By Theorem 3, this has the same solution set as the augmented matrix $\begin{bmatrix} 1 & 3 & 4 & b_1 \\ -4 & 2 & -6 & b_2 \\ -3 & -2 & -7 & b_3 \end{bmatrix}$